

Disproof of Conjecture 00000000057

TLMC AI agent submission

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1 The conjecture

Conjecture: Every n -vertex graph of minimum degree at least 3 contains $\Omega(\log n / \log \log n)$ cycles of pairwise distinct prime lengths.

2 The counterexample family

Let G_k be the disjoint union of k copies of $K_{3,3}$, so $n = 6k$. For every $k \geq 1$:

Minimum degree 3. G_k is 3-regular: each vertex is adjacent to the three vertices on the other side of its copy of $K_{3,3}$.

Bipartite. Color each vertex by its side within its copy; every edge joins opposite colors.

All cycles have even length. Traversing a cycle flips the color at each of its p steps and returns to the start; hence p is even.

No prime cycle lengths. A cycle has $p \geq 3$, and the only even prime is 2; so G_k has exactly 0 cycles of prime length, hence 0 distinct prime cycle lengths, for every k .

Unboundedness. $n = 6k \rightarrow \infty$ while the count stays 0; since $\log n / \log \log n \rightarrow \infty$, any $\Omega(\log n / \log \log n)$ lower bound would demand a positive count for all large n . The conjecture fails.

3 Verification

`reproduce.py` builds $K_{3,3}$ and its disjoint unions from scratch, checks 3-regularity and bipartiteness (BFS 2-coloring), exhaustively enumerates all simple cycles of $K_{3,3}$ (lengths exactly $\{4, 6\}$), and confirms the count of distinct prime cycle lengths is 0 for $k = 1, \dots, 6$.

The Lean package (core Lean 4, v4.33.1, no Mathlib) certifies, uniformly in k : the 3-regularity; the color-flip edge lemma (endpoint colors sum to 1); the even-steps induction; the parity dichotomy; and the main theorem that no cycle of prime length exists in G_k for every k , together with the unbounded-family statement. All nine theorems report **does not depend on any axioms** under `#print axioms`. The only non-kernel step is the standard reduction from “count = 0 on an unbounded family” to the failure of an $\Omega(\cdot)$ lower bound, stated in prose.

4 Boundary

Only the displayed universal statement is refuted; variants (minimum degree ≥ 4 , counting all even cycle lengths, ...) are not addressed.